SR UNIVERSITY

SCHOOL OF CS&AI — DEPARTMENT OF CSE

SOFTWARE ENGINEERING AND SYSTEM DESIGN (24CS301PC311)

ASSIGNMENT 05 / 12: AGILE / SCRUM SIMULATION REPORT

Case Study 1: Online Food Ordering Application

Student Full Name	**M. Hemavardhan**
Hall Ticket Number	**2403A51422**
Course Code & Title	24CS301PC311 — Software Engineering and System Design
Academic Year & Semester	2026–2027 | III Year / I Semester (III/I SEM)
Curriculum Regulation	R24
Department & School	Dept. of CSE | School of CS&AI | SR University
Course Coordinator	Dr. N. Sharmila Banu
Course Instructor	Dr. Asha S
GitHub Repository	`https://github.com/Mhemavardhan/2403a51422-sesd-assignment-5`
GitHub Projects Board	`https://github.com/users/Mhemavardhan/projects/1`

---

1. Executive Summary & Problem Statement

This assignment report delivers a comprehensive Agile/Scrum simulation for **Case Study 1: Online Food Ordering App** from the official SESD Assignment 5 specification. The project designs and delivers a full-stack solution called **FoodSwift**, allowing campus students and faculty to browse restaurants, explore categorized menus, manage a shopping cart, simulate digital payments, and track orders through a live 5-stage progression stepper.

2. Product Backlog & User Stories

Using Planning Poker relative estimation with the Fibonacci sequence (1, 2, 3, 5, 8), the Product Backlog was partitioned into 5 core user stories and 12 sub-issues totaling 26 Story Points, directly recreating the reference assignment workflow.

Issue #	Type	Title / User Story	Priority	Points	Iteration	Status
#1	Epic	**Customer Login and Registration**	High	3	Iteration 1	In Progress
↳ #7	Task	Create Login UI	High	1	Iteration 1	In Progress
↳ #8	Task	Create Username Field	High	1	Iteration 1	Todo
↳ #9	Task	Create Password Field	High	1	Iteration 1	Todo
↳ #10	Task	Implement Login Validation	High	1	Iteration 1	Testing
↳ #11	Task	Display Error Message	Medium	1	Iteration 1	Todo
↳ #12	Task	Test Login	High	1	Iteration 1	Todo
#3	Epic	**Browse Restaurants and Menus**	High	5	Iteration 1	Todo
↳ #13	Task	Create Restaurant Page	High	1	Iteration 1	Todo
↳ #14	Task	Display Restaurants	High	1	Iteration 1	Todo
↳ #15	Task	Create Menu Page	High	1	Iteration 1	Todo
↳ #16	Task	Display Food Items	High	1	Iteration 1	Todo
↳ #17	Task	Display Food Prices	Medium	1	Iteration 1	Todo
↳ #18	Task	Test Menu	High	1	Iteration 1	Todo
#4	Epic	**Shopping Cart**	High	5	Iteration 2	Todo
#5	Epic	**Online Payment (Mock Gateway)**	High	8	Iteration 2	Done

Issue #	Type	Title / User Story	Priority	Points	Iteration	Status
#6	Epic	**Order Tracking**	Medium	5	Iteration 3	Todo

3. Sprint Schedules & Increments

- **Iteration 1 (Oct 14 – Oct 27):** Foundation sprint delivering customer authentication with salted SHA-256 hashing and restaurant/menu browsing.
- **Iteration 2 (Oct 28 – Nov 10):** Ordering sprint delivering dynamic shopping cart management and mock digital payment processing.
- **Iteration 3 (Nov 11 – Nov 24):** Quality & tracking sprint delivering real-time 5-stage order status tracking, automated test verification, and documentation.

4. Automated Test Verification

All automated unit and integration tests executed cleanly with 100% pass rate in 0.008 seconds:

```
test_01_database_tables_exist: PASS
test_02_password_hashing_security: PASS (Salted SHA-256 confirmed)
test_03_default_seeded_restaurants: PASS (4 restaurants, 16 menu items)
test_04_user_registration_and_authentication: PASS
test_05_cart_and_mock_checkout: PASS (Subtotal, delivery, mock payment verified)
test_06_order_tracking_progression: PASS (5 stages verified: Placed -> Delivered)

Total: 6 Passed, 0 Failed. Status: 100% OK
```

5. Student Verification & Sign-Off

Student Name: M. Hemavardhan
Hall Ticket Number: 2403A51422
Subject: Software Engineering and System Design (SESD)
Institution: SR University, Warangal